

Scientific Communities in the Literature Sample

Qualitative category development from article titles and keywords

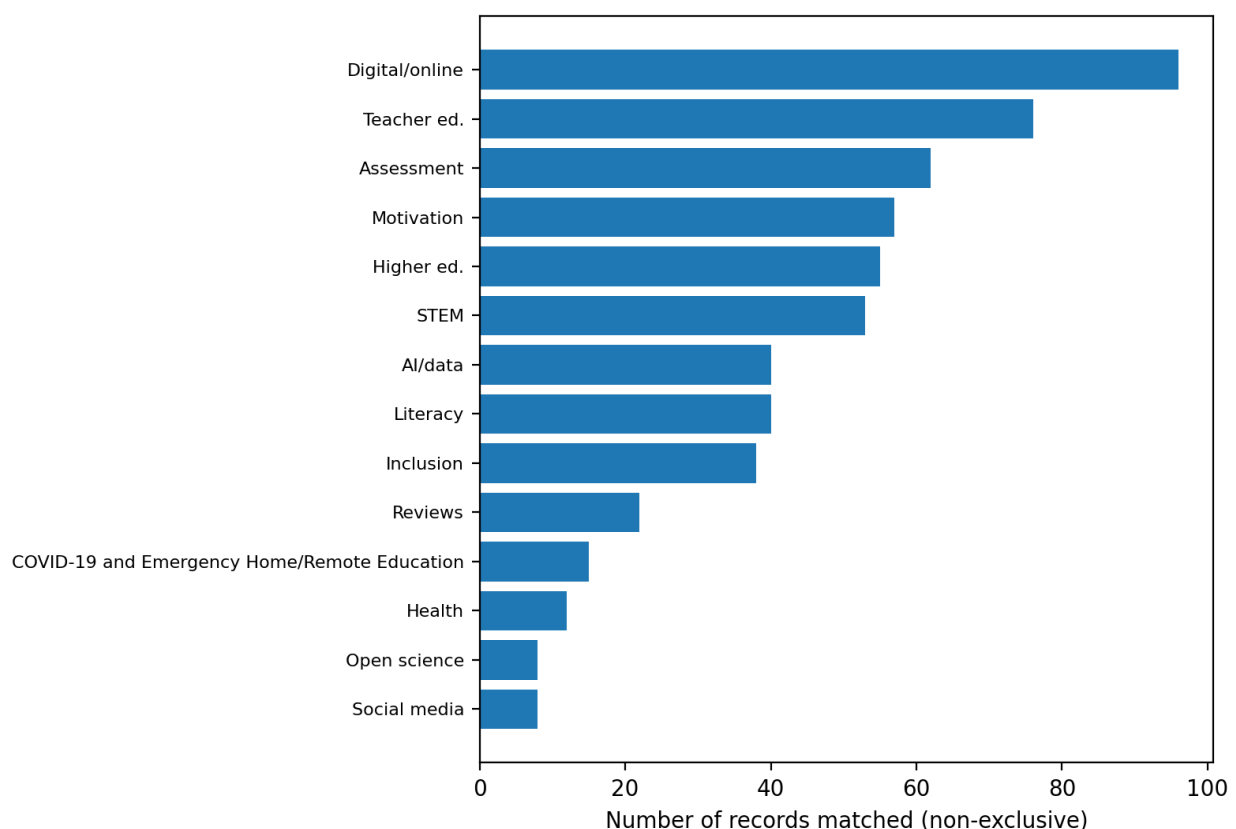
Source file: data_with_links.csv. Unit of analysis: 250 literature records. Fields used: **title** and **keywords**. The categories below are inductive content categories intended to identify scientific communities represented in the sample.

Methodological note

I combined each record's title and keywords, reviewed recurring terms and clusters, and formed thematic community categories. Counts are **non-exclusive**: one record can be counted in several communities when its title or keywords indicate overlap, for example AI in STEM education or online teacher professional development. The analysis is therefore a map of thematic presence, not a mutually exclusive classification of papers.

Because the coding uses only titles and keywords, the categories should be read as a high-level qualitative coding scheme. Full-text coding would allow finer differentiation of borderline cases.

Overview of category prevalence



Category list with definitions and examples

Category	Definition	Indicative count	Example from sample
Digital, Online, and Technology-Enhanced Learning	Research communities studying digital learning environments, educational technologies, online courses, virtual or augmented reality, games, and platform-supported learning processes.	96 / 250 (38.4%)	<i>Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database</i> Keywords: ICT; mathematics education; bibliometric analysis; Scopus

Category	Definition	Indicative count	Example from sample
Teacher Education, Professional Development, and Classroom Practice	Work focused on teachers, teacher educators, professional learning, instructional design, classroom management, curriculum implementation, and classroom practice.	76 / 250 (30.4%)	<i>A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic</i> Keywords: online spaces; teacher learning; professional learning networks; COVID-19; social media; stress
Assessment, Psychometrics, and Measurement	Communities concerned with tests, scales, instruments, psychometric validation, large-scale assessments, score interpretation, and classification or evaluation methods.	62 / 250 (24.8%)	<i>Exploring the Associations Between MAP Growth and SBA by Demographics</i> Keywords: formative assessment; summative assessment; measures of academic progress; smarter balanced assessment consortium; growth
Motivation, Well-Being, and Socio-Emotional Development	Studies of affective, motivational, psychological, identity-related, and social-emotional dimensions of learning and educational participation.	57 / 250 (22.8%)	<i>Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources</i> Keywords: COVID-19 pandemic; college students; psychological well-being; emotions; experience sampling method; personality; intraindividual trajectories
Higher Education, Doctoral Training, and Career Pathways	Communities examining universities, colleges, doctoral education, graduate training, admissions, academic progress, employability, and transitions into careers.	55 / 250 (22.0%)	<i>Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education</i> Keywords: graduate education; doctoral training; program evaluation; career outcomes; taxonomy; PhD
STEM, Mathematics, and Science Education	Subject-specific communities in science, technology, engineering, mathematics, computational education, and science-related outreach or pedagogy.	53 / 250 (21.2%)	<i>Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database</i> Keywords: ICT; mathematics education; bibliometric analysis; Scopus
AI, Learning Analytics, and Educational Data Science	Communities using or studying artificial intelligence, machine learning, NLP, educational data science, learning analytics, automated scoring, prediction, and generative AI.	40 / 250 (16.0%)	<i>Understanding Public Sentiment About Educational Reforms: The Next Generation Science Standards on Twitter</i> Keywords: public sentiment, Next Generation Science Standards, sentiment analysis, educational data science
Literacy, Language, and Reading Education	Communities focused on reading, writing, language learning, literacy integration, vocabulary, comprehension, textbooks, and language-related interventions or assessments.	40 / 250 (16.0%)	<i>Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks</i> Keywords: artificial intelligence; case studies; content analysis; curriculum; data science; gender studies; history; natural language processing; race; textbooks; textual analysis
Inclusion, Special Education, and Equity	Research addressing inclusive education, special educational needs, disability, rural access, equity gaps, underrepresented groups, diversity, and social justice in education.	38 / 250 (15.2%)	<i>An Evaluation of the Educational Impact of College Campus Visits: A Randomized Experiment</i> Keywords: college access; experimental research; postsecondary education; program evaluation; rural education; socioemotional skills; survey research
Reviews, Bibliometrics, and Evidence Synthesis	Methodological and meta-research communities that map, synthesize, or quantify bodies of literature through reviews, meta-analyses, bibliometrics, or network analyses.	22 / 250 (8.8%)	<i>Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database</i> Keywords: ICT; mathematics education; bibliometric analysis; Scopus
COVID-19 and Emergency Home/Remote Education	Communities examining pandemic-related disruption, lockdown learning, home education, emergency online teaching, and post-pandemic recovery in education.	15 / 250 (6.0%)	<i>Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources</i> Keywords: COVID-19 pandemic; college students; psychological well-being; emotions; experience sampling method; personality; intraindividual trajectories

Category	Definition	Indicative count	Example from sample
Health, Nursing, and Physical Education	Research at the intersection of education with nursing, medical or health professions education, physical education, physical activity, and health-related quality of life.	12 / 250 (4.8%)	<i>Studying with GenAI: cross-sectional study on usage patterns, needs, competencies, and ethical perspectives of medical informatics students</i> Keywords: generative AI; GenAI literacy; GenAI usage patterns; higher education; ethical attitudes; GenAI competences and skills; DigComp 2.2; GenAI skill development
Open Science, OER, and Research Infrastructure	Communities concerned with open education, open educational resources, open science, data sharing, reproducibility, preregistration, shared research infrastructure, and credibility.	8 / 250 (3.2%)	<i>Open Education Science</i> Keywords: preregistration; open science; registered report; open access
Social Media, Networks, and Informal Learning Spaces	Communities studying education-related social media, online affinity spaces, professional networks, social network analysis, public sentiment, and informal or networked learning.	8 / 250 (3.2%)	<i>A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic</i> Keywords: online spaces; teacher learning; professional learning networks; COVID-19; social media; stress

Interpretive summary

The strongest communities in this sample are digital or online learning, teacher education and professional development, assessment and psychometrics, motivation/well-being, higher education, STEM education, and AI/educational data science. This suggests a literature selection centered on technology-mediated education, measurement/evaluation, and learner or teacher development.

Several communities are strongly interconnected. For example, AI and data science often appear together with assessment, STEM education, or online learning. Teacher education overlaps with digital learning through professional development around technology, VR/AR, or online professional learning networks. Open science appears as a smaller but distinct community, focused on open education science, open resources, data sharing, and credibility.

Recommended use of the categories

For subsequent manual coding, these categories can be used as a multi-label codebook. A paper should receive all community codes that are explicitly signaled by the title or keywords. If a single primary community is required, choose the community that best captures the main object of study rather than the method. For example, a machine-learning paper about reading assessment could receive the multi-label codes AI/data science, literacy, and assessment; as a primary code, it would likely be assessment or literacy depending on the research question.

Compact codebook

Digital, Online, and Technology-Enhanced Learning

Definition: Research communities studying digital learning environments, educational technologies, online courses, virtual or augmented reality, games, and platform-supported learning processes.

Example: *Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database*

Teacher Education, Professional Development, and Classroom Practice

Definition: Work focused on teachers, teacher educators, professional learning, instructional design, classroom management, curriculum implementation, and classroom practice.

Example: *A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic*

Assessment, Psychometrics, and Measurement

Definition: Communities concerned with tests, scales, instruments, psychometric validation, large-scale assessments, score interpretation, and classification or evaluation methods.

Example: *Exploring the Associations Between MAP Growth and SBA by Demographics*

Motivation, Well-Being, and Socio-Emotional Development

Definition: Studies of affective, motivational, psychological, identity-related, and social-emotional dimensions of learning and educational participation.

Example: *Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources*

Higher Education, Doctoral Training, and Career Pathways

Definition: Communities examining universities, colleges, doctoral education, graduate training, admissions, academic progress, employability, and transitions into careers.

Example: *Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education*

STEM, Mathematics, and Science Education

Definition: Subject-specific communities in science, technology, engineering, mathematics, computational education, and science-related outreach or pedagogy.

Example: *Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database*

AI, Learning Analytics, and Educational Data Science

Definition: Communities using or studying artificial intelligence, machine learning, NLP, educational data science, learning analytics, automated scoring, prediction, and generative AI.

Example: *Understanding Public Sentiment About Educational Reforms: The Next Generation Science Standards on Twitter*

Literacy, Language, and Reading Education

Definition: Communities focused on reading, writing, language learning, literacy integration, vocabulary, comprehension, textbooks, and language-related interventions or assessments.

Example: *Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks*

Inclusion, Special Education, and Equity

Definition: Research addressing inclusive education, special educational needs, disability, rural access, equity gaps, underrepresented groups, diversity, and social justice in education.

Example: *An Evaluation of the Educational Impact of College Campus Visits: A Randomized Experiment*

Reviews, Bibliometrics, and Evidence Synthesis

Definition: Methodological and meta-research communities that map, synthesize, or quantify bodies of literature through reviews, meta-analyses, bibliometrics, or network analyses.

Example: *Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database*

COVID-19 and Emergency Home/Remote Education

Definition: Communities examining pandemic-related disruption, lockdown learning, home education, emergency online teaching, and post-pandemic recovery in education.

Example: *Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources*

Health, Nursing, and Physical Education

Definition: Research at the intersection of education with nursing, medical or health professions education, physical education, physical activity, and health-related quality of life.

Example: *Studying with GenAI: cross-sectional study on usage patterns, needs, competencies, and ethical perspectives of medical informatics students*

Open Science, OER, and Research Infrastructure

Definition: Communities concerned with open education, open educational resources, open science, data sharing, reproducibility, preregistration, shared research infrastructure, and credibility.

Example: *Open Education Science*

Social Media, Networks, and Informal Learning Spaces

Definition: Communities studying education-related social media, online affinity spaces, professional networks, social network analysis, public sentiment, and informal or networked learning.

Example: *A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic*